Lecture 4: Program Development life cycle(PDLC)

Lesson objectives:

By the end of this lesson, the learner should be able to:

- Describe the program development life cycle stages
- Describe the tools used in program design.
- Describe the properties of an algorithm.
- Convert an algorithm to a program.

Program Development life cycle(PDLC)

Set of steps for creating a program

a) Problem specification:

- Establish the **processes** involved in a given scenario.
- Find the gap- what's not working as expected
- What the program will do.

b) Problem analysis: Break the problem into smaller components.

- Analyzing the problem will eventually make it easier to solve
- Problem- What to do
- The inputs- Values to be processed
- The process- How inputs will be converted into output
- Output- The expected result/outcome

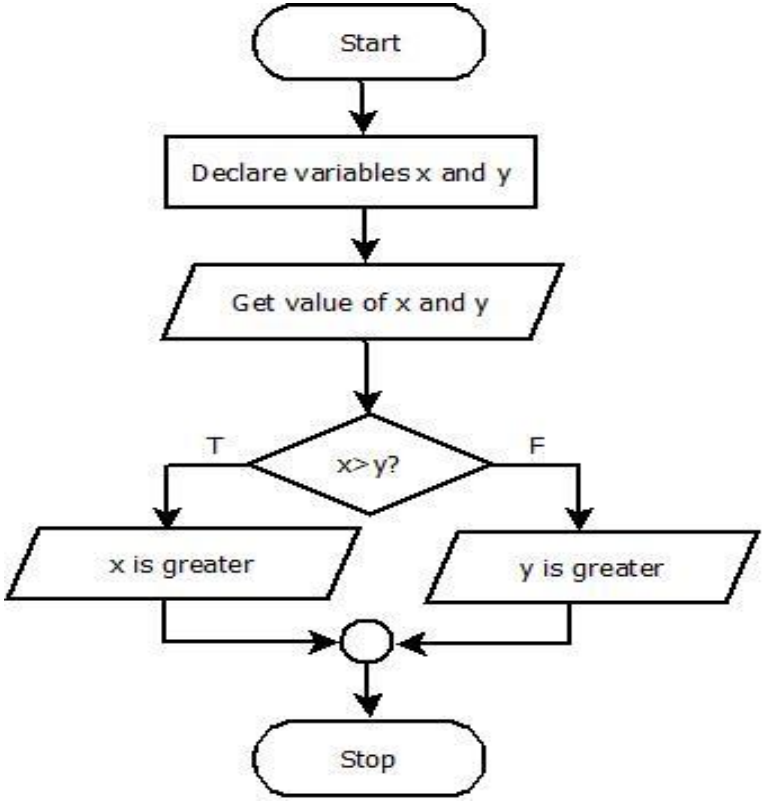
Program Development life cycle(PDLC)

c) Program design

- Understanding the purpose of a program usually involves understanding its **IPO:-**
  - **Inputs**
  - **Processing**
  - **Outputs**
- The IPO approach is useful for teaching/learning programming**.**
- Sometimes, it might help to visualize the program running on the computer.
- Programmers are expected to be able to understand and do **flowcharting** and **pseudo code**.
- The program design(algorithm) is always done before coding.
- Algorithm is a set of steps for solving a problem.
- A good algorithm should be:-
  i. Completeness-Non ambiguous – clarity.
  ii. Finiteness – Activities must come to an end(start and finish)
  iii. Efficiency- Ability to execute task using minimum amount of resource.
  iv. Effectiveness- Ability to solve the problem

Program Development life cycle(PDLC)

Program design using flowchart



Advantages:

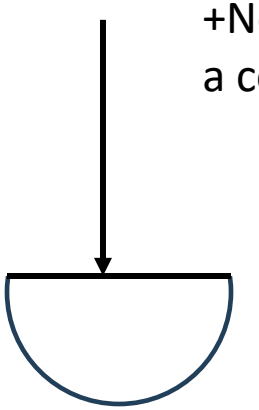
+ Represent the big picture i.e.logic

+Easy to understand – a picture is worth 1000 words

Disadvantage:

+Can be complex if the solution is detailed

+Not easy to translate into a code



Program Development life cycle(PDLC)

Program design using pseudo code

```
1. START
2. Declare variables x and y
3. Get value of x and y
4. IF x>y?
        Display x
5. ELSE
        Display y
6. STOP
```

Advantages:

+ Resembles the code
+ Easy to understand
+Represent details in the design.
+Easy to translate into code

Disadvantage:

+Language can be ambiguous
+Does not give the bigger picture

Program Development life cycle(PDLC)

Program design using pseudo code

4. IF X>Y>?
    4.1. Calculate Z=X+Y
    4.2. Display Z
5. ELSE
    5.1. Calculate Z=X-Y
    5.2. Display Z
6. …..

Advantages:

+ Resembles the code
+ Easy to understand
+Represent details in the design.
+Easy to translate into code

Disadvantage:

+Language can be ambiguous
+Does not give the bigger picture

Program Development life cycle(PDLC)

Set of steps for creating a program

d) Coding: Translate the design into executable instructions.

- Select appropriate programming tools
- Follow the syntax and standard conventions
- Comment statements to explain
- Ensure the code is kept simple and readable

Program Development life cycle(PDLC)

d) Coding using C

```c
#include<stdio.h>
int main(){
    int x,y;
    printf("Enter valueb of x and y\n");
    scanf("%d%d",&x,&y);
    if(x>y){
        printf("X is greater\n");
    }
    else{
        printf("y is greater\n");
    }
    return 0;
}
```

Program Development life cycle(PDLC)

Set of steps for creating a program

e) Testing and debugging: Run the program to identify and remove.

- Test can be done for module/function.
- Testing can also be done after combining/integrating separate modules.
- Testing can also be done by users(black box testing)
- Testing can also be done by the programmers(white box testing)

Program Development life cycle(PDLC)

Set of steps for creating a program

f) Documentation: Create external documentation(user manuals) and internal documentation(comments).

- User manuals need to be provide to users showing basic operations to the program.
- The user manual can manual or electronic.
- Inside the code, documentation is done using comments(brief explanations)

Program Development life cycle(PDLC)

Set of steps for creating a program

Internal documentation(comments)

```
//To include the header file
#include<stdio.h>
//start program
int main()
{
//Declare variables
int x,y;
//read the numbers
printf("Enter first number\n");
scanf("%d",&x);
printf("Enter second number\n");
scanf("%d",&y);
//test for the greater number
if(x>y){
printf("%d",x);
}
else{
printf("%d",y);
}
//end
return 0;
}
```

Program Development life cycle(PDLC)

Set of steps for creating a program

g). Maintenance: Aim at ensuring the program continues to perform optimally.

- Update the code
- Repair the code
- Tune the code

Program Development life cycle(PDLC)

Activity 4:

A retailer has requested you to help him/her come up with an application to:

- Read product quantity sold and its unit price.
- Give a discount of 2% if the product cost above 3000 KES.
- Product cost is calculated as **quantity x unit price**

Required:

Design a program for the above case using a flowchart/pseudo code

Program Development life cycle(PDLC)

Exercise 4:

a) Describe **three** characteristics of a good algorithm

b) Briefly describe the steps followed in program development life cycle.

c) Identify and explain the components of a flowchart.

d) Design a program using pseudocode to receive two numbers store them in variables X1 and X2, then swap the two numbers(without a third variable) and display the swapped numbers.

Reference:

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THANK YOU